Reza Baharani

Senior AI Engineer | Health Sensing & On-Device ML | PhD

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Google Scholar

Experience

ForesightCares Inc., Senior AI Engineer — Charlotte, NC — Jan 2022 – present — 4 years 8 months

- Architected and deployed real-time 3D human pose estimation pipeline on iPad achieving 30fps inference for clinical mobility assessments (CDC STEADI: TUG, 30-Second Chair Stand, 4-Stage Balance)
- Engineered on-device AI inference system using CoreML and SceneKit, eliminating cloud dependencies and reducing model bundle size by migrating from third-party ML frameworks to native Apple infrastructure
- Designed multi-stage signal processing pipeline (One Euro filters, Butterworth IIR, median filters) achieving clinical-grade measurement accuracy for joint angle computation and gait analysis
- Built multi-agent AI orchestration system automating regulatory compliance analysis for CMS CY 2026 Medicare RTM billing codes, producing audit-ready documentation, workbooks, and presentations
- Built serverless AI-powered exercise recommendation service on AWS Lambda, leveraging cloud inference and Jinja2-based prompt engineering to deliver personalized mobility programs from a curated library of 41 exercises based on CDC STEADI fall-risk protocols
- Developed Field-of-View calibration model to correct 3D pose estimation across varying camera angles, improving measurement accuracy for at-home and clinical deployment scenarios
- Implemented state machine architecture for 5+ clinical assessment protocols with hysteresis-based transitions, handling edge cases including camera occlusion and multi-person environments
- Co-invented AI-powered mobility assessment system (US Patent 12,343,138 B2), enabling automated fall risk evaluation and real-time exercise coaching via 3D skeleton modeling from monocular video
- Created 3D motion capture dataset using IMU-based mocap system for training and validation, collecting synchronized 60fps video and 400Hz motion data from 100+ subjects to validate pose estimation accuracy

UNC Charlotte, Research Scientist — Charlotte, NC — Oct 2024 – Mar 2025 — 6 months

- Developed self-supervised human motion foundation model using discrete motion tokenization and Vision Transformer pretraining, achieving competitive action classification (89.4%), one-shot recognition (57.4%), and anomaly detection (76.56% AUC-ROC) with a single unified backbone

UNC Charlotte, Postdoctoral Researcher — Charlotte, NC — Aug 2021 – Dec 2022 — 1 year 5 months

- Architected scalable intelligent video surveillance systems for AIoT, including Ancilia and REVAMP2T, enabling privacy-aware multi-camera pedestrian tracking with real-time edge inference across distributed infrastructure (IEEE IoT Journal; 89 and 93 citations)

UNC Charlotte, Graduate Research Assistant — Charlotte, NC — Aug 2017 – Aug 2021 — 4 years 1 month

- Designed algorithm/architecture co-design framework (DeepDive) for deep separable CNNs on Xilinx FPGA, enabling power-efficient edge inference for MobileNetV2 (published at GLSVLSI 2021)
- Developed LSTM-based real-time reliability prediction system for power electronics converters, achieving scalable monitoring on IoT edge devices at 1.87W power budget (published in IEEE IoT Journal, 74 citations)
- Created lightweight temporal convolutional network for vehicle trajectory prediction in highways, matching state-of-the-art accuracy with significantly reduced computational complexity (published in IEEE TITS, 97 citations)
- Co-developed gem5-SALAM system architecture simulator for LLVM-based accelerator modeling (74 citations, IEEE/ACM MICRO 2020)
- Published 12+ peer-reviewed papers across IEEE, ACM venues accumulating 645+ citations (h-index: 11) spanning edge AI, FPGA acceleration, intelligent transportation, and IoT
- Designed attention-based temporal convolutional networks (ATCN) enabling fast time-series classification on resource-constrained embedded devices with formalized hyper-parameters for application-specific architecture tuning (published in ACM TECS)
- Deployed ATCN on Cortex-M7 microcontroller consuming only 49% of 320KB RAM and 15% of 1MB flash; validated across 70 UCR 2018 time-series benchmarks using data augmentation (jittering, magnitude warping, window warping, scaling)
- Developed real-time person re-identification system at the edge using mixed precision (FP16/FP32) inference, achieving efficient tracking on embedded GPUs with MobileNetV2 backbone (published at ICIAR 2019, 17 citations)

Education

PhD **University of North Carolina at Charlotte**, Electrical and Computer Engineering — Charlotte, NC — Aug 2017 – Aug 2021

- Advisor: Prof. Hamed Tabkhi
- Focus: Computer Architecture and Deep Learning

Publications

MoFM: A Large-Scale Human Motion Foundation Model — Feb 2025
Reza Baharani, G.A. Noghre, A.D. Pazho, G. Maldonado, H. Tabkhi
(arXiv:2502.05432)

DeepTrack: Lightweight Deep Learning for Vehicle Trajectory Prediction in Highways — Jan 2022
V. Katariya, *Reza Baharani*, et al.
(IEEE Trans. Intelligent Transportation Systems)

REVAMP2T: Real-Time Edge Video Analytics for Multicamera Privacy-Aware Pedestrian Tracking — Jan 2019
C. Neff, M. Mendieta, S. Mohan, *Reza Baharani*, et al.
(IEEE IoT Journal)

Ancilia: Scalable Intelligent Video Surveillance for the AIoT — Jan 2023
A.D. Pazho, C. Neff, G.A. Noghre, B.R. Ardabili, S. Yao, *Reza Baharani*, H. Tabkhi
(IEEE IoT Journal)

Real-Time Deep Learning at the Edge for Scalable Reliability Modeling of Si-MOSFET Power Electronics Converters — Jan 2019
Reza Baharani, et al.
(IEEE IoT Journal)

ATCN: Resource-Efficient Processing of Time Series on Edge — Jan 2022
Reza Baharani, H. Tabkhi
(ACM Trans. Embedded Computing Systems (TECS))

Patents

- US 12,343,138 B2: AI Powered Mobility Assessment System, ForesightCares Inc., Jul. 2025. Inventors: H. Tabkhivayghan, M. Azarbayjani, R. Baharani.

Skills

Languages: Python, C/C++, Java, Swift, CUDA

ML/DL Frameworks: PyTorch, TensorFlow, CoreML, ONNX, Vision Transformers

Apple/Mobile: SwiftUI, SceneKit, Combine, CoreML, iOS/iPadOS

DevOps & Cloud: Git, Docker, AWS (Lambda, S3, DynamoDB), Terraform, ROS 2

Edge & Embedded: NVIDIA Jetson, Xilinx FPGA, ARM Cortex-M7, Mixed Precision (FP16/FP32)

Signal Processing: One Euro Filters, Butterworth IIR, Median Filters, Kalman Filtering

Domains: Computer Vision, 3D Pose Estimation, Human Motion Understanding, Self-Supervised Learning, Embedded AI, Signal Processing